Midnight Buildathon · AKINDO Wave 2

🌙 NightPoll

Vote once. Stay anonymous. Prove the tally.

Anonymous community polls on Midnight: zero-knowledge membership proofs + one-time nullifiers.
Only the tally is public.

hycorange · `github.com/hycorange/nightpoll` · Apache-2.0

PROBLEM

Polls are accountable or private, rarely both

Accountable, not private

You log in, so the operator (or the chain) sees exactly who voted for what. People self-censor on sensitive questions.

Private, not accountable

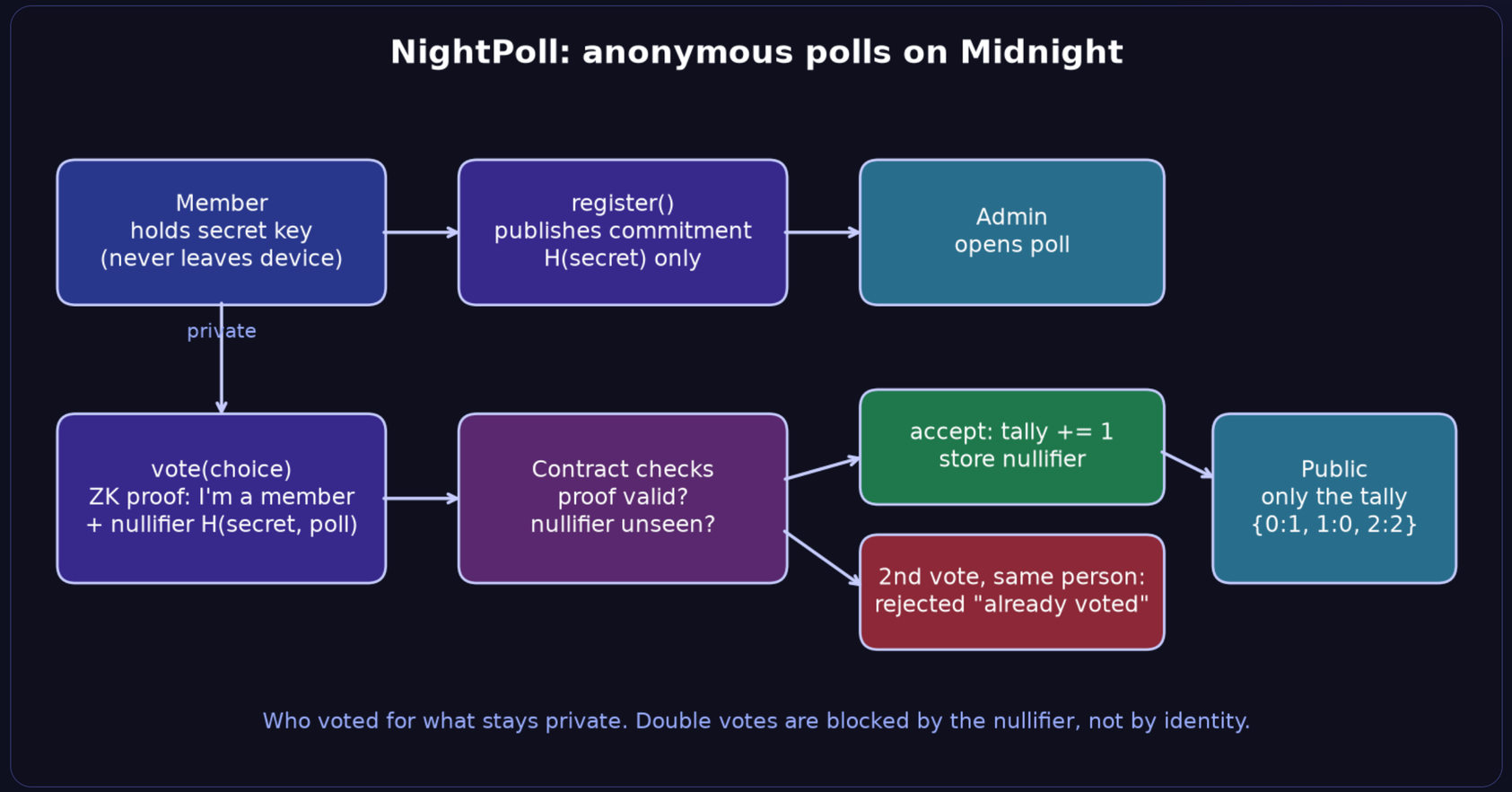
Anonymous forms can't stop ballot stuffing, and you have to trust whoever counts.

Where it hurts:

DAO treasury votes | Workplace pulse surveys | Union votes | Moderation decisions

SOLUTION

Commitment → ZK proof → nullifier → tally



Eligibility without identity · **one person, one vote** (same key ⇒ same nullifier ⇒ rejected) · **verifiable tally**

WHY MIDNIGHT

Private state + public ledger, in one contract

PUBLIC (ON CHAIN)	PRIVATE (ON DEVICE)
question, option count, phase, admin key hash	each member's secret key
member commitments (opaque hashes)	which commitment is whose
nullifiers (unlinkable to commitments)	which member cast which vote
per-option tally, total votes	the Merkle path used in each proof

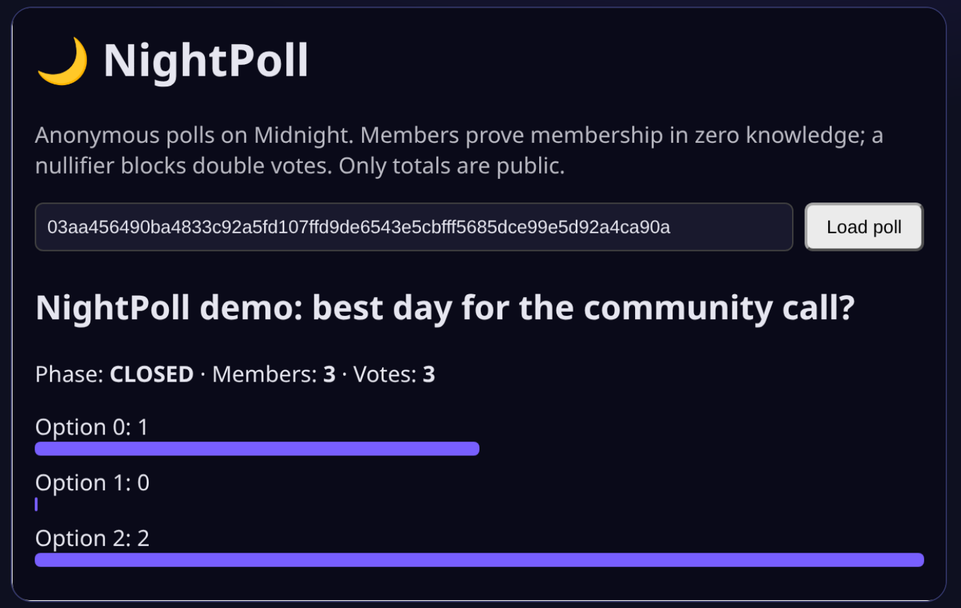
- Circuit logic runs locally over **private state**; the chain verifies a ZK proof of that run.
- `vote()` discloses only the **nullifier** and the **choice**.
- Commitment and nullifier use different domain tags, so a vote can't be linked to a registration.
- Compact keeps the circuit short and readable.

DEMO · MIDNIGHT PREVIEW TESTNET · 2026-10-11 UTC

Every step is a real Preview transaction

`contract 03aa456490ba4833c92a5fd107ffd9de6543e5cbfff5685dce99e5d92a4ca90a`

STEP	TX HASH	BLOCK
Deploy poll (3 options)	`99bd34ed5c0cf769…`	1252343
alice self-registers	`a37f7565e3d76d3d…`	1252351
bob self-registers	`2cc1d773977b2a91…`	1252355
admin adds carol's commitment	`41b5e45d03286e92…`	1252360
admin opens voting	`7912582deb76e9df…`	1252365
anonymous vote #1 (opt 0)	`40de0a3c90d8f39f…`	1252370
anonymous vote #2 (opt 2)	`088025fce8083534…`	1252374
anonymous vote #3 (opt 2)	`5e9f6d015f51675e…`	1252378
admin closes poll	`5035a46d487f369b…`	1252383
alice votes again	**rejected:** `NightPoll: already voted`	



Read-only viewer decoding the live ledger from the Preview indexer: **CLOSED** · 3 members · 3 votes · tally **1 / 0 / 2**

Explorer: preview.midnightexplorer.com/contracts/03aa4564…4ca90a

ENGINEERING & QA

Built and tested on real Midnight infrastructure

5 circuits

`register` · `addMember` · `openVoting` · `closePoll` · `vote`. Compact 0.31.1, compiles with ZK keys. `HistoricMerkleTree<10>` (up to 1024 members).

19 / 19 simulator tests

Including adversarial witnesses: forged Merkle path from another tree, borrowed path, double vote for a different option, non-member, out-of-range option, phase misuse.

Real-chain end-to-end

Same flow on a local `undeployed` network (official node 0.22.3, indexer 4.0.1, proof server 8.1.3, no Docker) and on the Preview testnet.

Two SDK pitfalls fixed

Duplicate onchain-runtime → `StateValue` error (pinned override). Stalled dust subscription → `InvalidDustSpendProof` (wait + wallet re-open).

THREAT MODEL & ROADMAP

Honest limits, concrete fixes

Known limits today

- **Fee-payer linkability:** one wallet paying for register + vote links them (the demo uses one wallet).
- **The choice is public** per vote tx (that's what keeps the tally verifiable); only the voter is hidden.
- **Timing / small sets** can reveal voters; anonymity grows with the member set.
- **Admin** controls phases and curated membership, but can't vote for others or see ballots.

Wave 3

- **DUST fee sponsorship / relayer**: the voter's wallet never appears on chain.
- **Browser voting with Lace** (DApp connector), proofs generated locally.
- **Credential-gated registration** via selective disclosure.
- **Multi-question surveys** and ranked choice, results revealed after close.

ADOPTION

From protocol to product

Who

DAO governance · online communities · employee pulse surveys · unions · moderation votes

Path

Embeddable widget / SDK for one-person-one-vote checks → hosted poll page for DAOs and communities

Sustainability

Free open polls; paid sponsored-fee tiers for organisations

🌙 NightPoll: vote once, stay anonymous, prove the tally.

`github.com/hycorange/nightpoll`